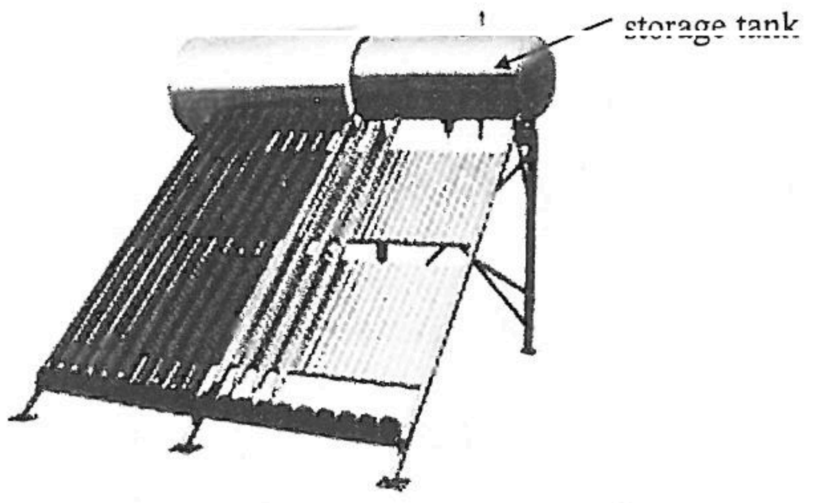
your answers in the spaces provided.

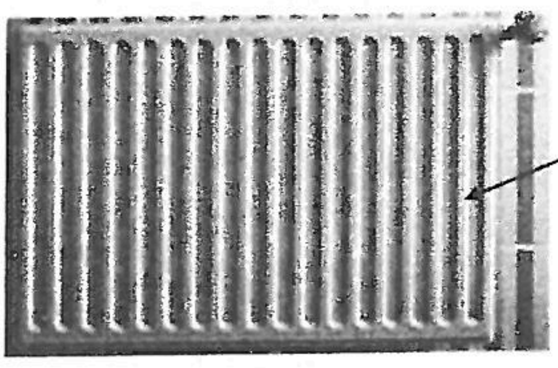
1.

Figure 1.1



A solar water heater shown in Figure 1.1 is installed on the rooftop of a house. During the day, the hea heats up 1.5 m^3 of water to 80 °C. At night, the hot water in the storage tank is circulated to the radiato (see Figure 1.2) in different rooms of the house to keep the rooms warm.

Figure 1.2



Given: density of water = 1000 kg m^{-3}
specific heat capacity of water = 4200 J kg^{-1} $°C^{-1}$

(a) Given that 15% of the energy is lost during the transfer of water, how much heat can be released fro the system to the rooms when the water temperature drops to 60 °C ? (3 mark

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(b) Given that during night time the radiators maintain an average output power of 4.5 kW, how long can the radiators maintain this average power until the water temperature in the system drops to 60 °C ? Give your answer in hours. (2 marks)

(c) The rate of heat released by the solar water heating system during the time period calculated in (b) is in fact not constant and gradually drops. Explain why this is so. (1 mark)

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1.

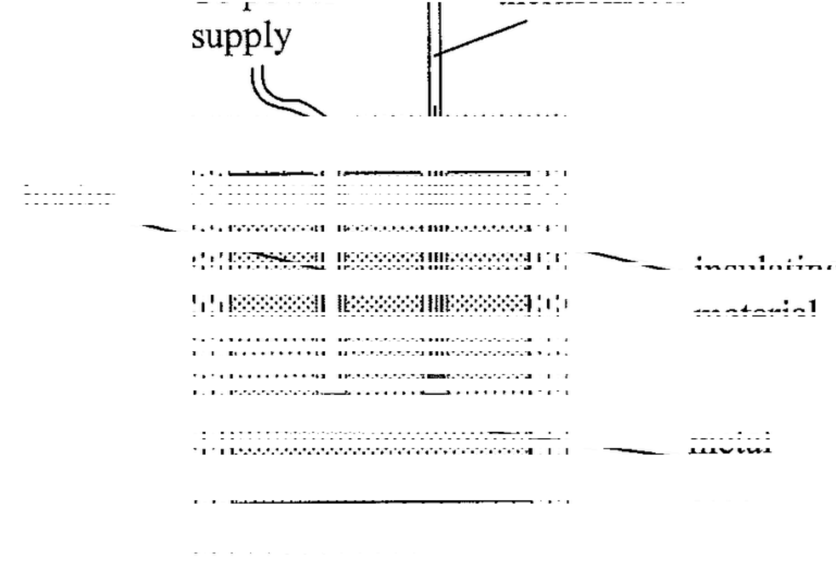
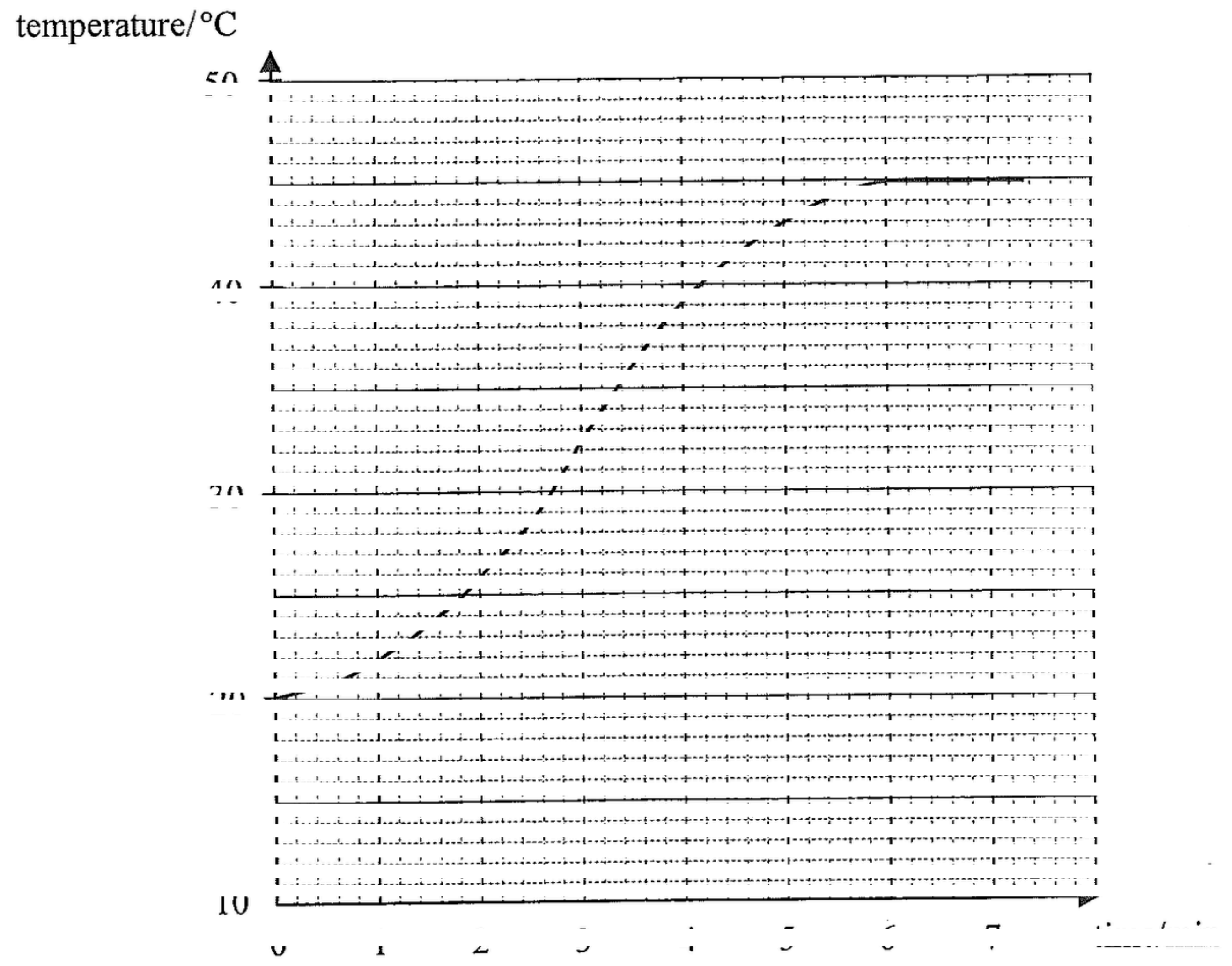


Figure 1.1

Figure 1.1 shows an experimental set-up to find the specific heat capacity of a metal. The metal block is wrapped by insulating material. A heater is connected to a power supply. It is switched on when the temperature of the metal block is 20 °C and then switched off when the temperature reaches 43 °C. The graph below shows the variation of the temperature of the metal block with time.



(a) Use the graph to find the duration that the heater is switched on. (1 mark)

(b) After the heater is switched off, the temperature of the metal block continues to rise for a while. Explain why. (1 mark)

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(c) Given: mass of the metal block = 0.80 kg
heater voltage = 12 V
heater current = 4.0 A

(i) By considering the maximum temperature rise of the metal block, calculate the specific heat capacity of the metal as found from the experiment. (2 marks)

(ii) Would your result be the same, higher, or lower than the actual value of the specific heat capacity of the metal ? Explain. (2 marks)

(d) This method is not suitable for measuring the specific heat capacity of a glass block. Explain.(1 mark)

1. The following experimental items are provided for estimating the specific heat capacity of bronze c_b:

a bronze sphere of mass 0.80 kg hung with a thread at room temperature T_0
a polystyrene cup containing 0.50 kg of water at room temperature T_0
a water bath maintained at 80 °C
a thermometer
a stirrer
a towel

(a) Describe the procedures of the experiment and state **TWO** experimental precautions to be taken. Write down an equation for finding c_b.
Given: specific heat capacity of water = 4200 J kg^{-1} °C^{-1} (6 marks)

(b) The value of c_b found in the experiment in (a) is lower than the actual value. Explain. (2 marks)

1. Figure 1.1 shows a domestic water heater. Tap water entering the heater is heated when it passes through the coil duct and exits as hot water at a certain temperature.

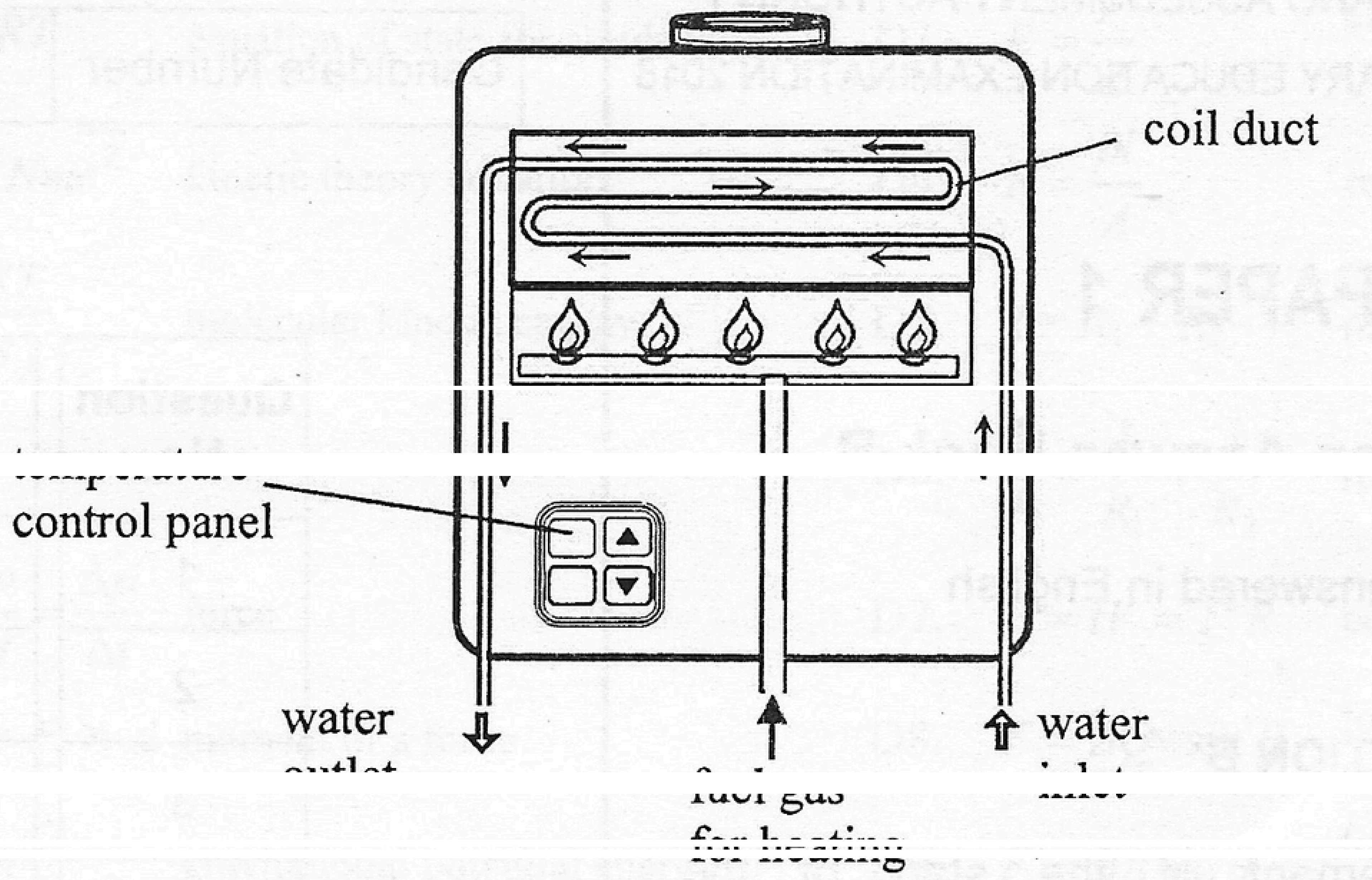

Figure 1.1

On a certain day in winter, the temperature of tap water is 15 °C. When the heater is in use, it delivers 6 kg of hot water at a temperature of 50 °C in 1 minute. Assume that there is no heat exchange between the heater and the surroundings. Given: specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

(a) Estimate the power supplied to the tap water by the heater. (3 marks)

(b) Assuming the power estimated in (a) remains unchanged, determine the flow rate of tap water into the heater, in kg per minute, such that hot water at a temperature of 40 °C is delivered by the heater. (2 marks)

1. In a restaurant, 'wontons in soup' is prepared by putting 5 pieces of cooked wonton at 4 °C into a bowl with 0.60 kg of soup at temperature 96 °C.

Given: average mass of each piece of wonton = 0.02 kg

specific heat capacity of wonton = 3300 J kg^{-1} °C^{-1}

specific heat capacity of soup = 4200 J kg^{-1} °C^{-1}

(a) Find the final temperature of the mixture. Assume that the heat capacity of the bowl and the heat loss to the surroundings are negligible. (2 marks)

(b) The soup in (a) is taken from a metallic container of heat capacity 2000 J °C^{-1} containing 16 kg of soup maintained at 96 °C by an immersion heater.

(i) Why does that energy have to be supplied by the heater to keep the soup at 96 °C ? (1 mark)

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(ii) A student used the following method to find the heater's operating power P: remove the heater from the container and record the temperature of the 16 kg of soup after 10 minutes. It is found that the temperature has dropped 9 °C. Estimate P. (3 marks)

(iii) If the student repeats the measurement after another 10 minutes, would the corresponding temperature drop be larger than, equal to or smaller than 9 °C ? Explain. (2 marks)

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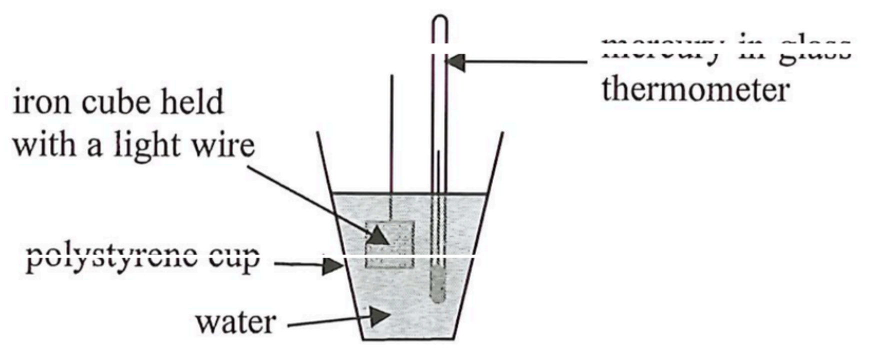
1. An iron cube of mass 200 g is heated to a few hundred degree Celsius (say at T °C). It is then quickly transferred into a polystyrene cup containing 600 g of water. The cube is totally immersed in the water as shown in Figure 1.1. The temperature of the heated iron cube can be estimated in this experiment.

Figure 1.1



Assume that no heat is lost to the surroundings and neglect the heat capacity of the polystyrene cup.

(a) The temperature of water recorded by the thermometer rises from 25 °C to 33 °C. Estimate T.
Given: The specific heat capacities of iron and water are 450 J kg^{-1} °C^{-1} and 4200 J kg^{-1} °C^{-1} respectively. (3 marks)

(b) What would happen to the water when this iron cube of high temperature just touches the water surface ? (1 mark)

(c) Explain why the actual temperature of the iron cube should be higher than that estimated in (a). (2 marks)

(d) (i) A mercury-in-glass thermometer is not suitable for measuring the temperature of the heated iron cube **directly**. Why ? (1 mark)

(ii) Suggest a kind of thermometer that can directly measure the temperature of this heated cube. (1 mark)

1. An empty glass cup is placed on a table under room temperature of 23 °C. 400 g of water at 100 °C is pour
into the cup. After a moment, the temperature of water in the cup is found to be 93 °C.

(a) State a cause for the drop of water temperature other than heat loss to the surrounding air. (1 mar

(b) Estimate the heat capacity H, in J °C^{-1}, of the glass cup. (2 mark
Given: specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

(c) (i) What is the kind of electromagnetic waves, other than visible light, mainly emitted from this cup of h
water ? (1 mar

(ii) Name a device in daily life which makes use of this kind of electromagnetic waves. (1 mar